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Bio-Diesel Preparation and Application in Linear Gel Through Efficient Hydrofracturing

V. Reddi and R. Bukahmseen, TAQA, Dammam, KSA

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Abstract

Biodiesel is produced from vegetable oil or animal fats. It is produced from triglyceride (oil) in the presence of methanol with a catalyst through a trans-esterification reaction, glycerol and methyl ester (Biodiesel) are obtained. This study describes the lab test methodology adopted to develop new hydrofracturing fluids, which is a bio-diesel linear gel, and will evaluate its performance efficiency in comparison with the conventional fossil diesel linear gel.

The lab test was conducted in accordance with API standards. Hydrofracturing Linear gel is prepared by treating Biodiesel with various water samples collected from different water bodies. The findings that were presented suggest that Biodiesel, an alternative eco-friendly fuel synthesized from "Jatropha seed". This research work explored to development of alternate methods for hydrofracturing processes, which are feasible and eco-friendly. The reaction mechanism of one molecule of ammonium persulfate yields two free radicals per reaction. The free radicals are released by the persulphate reaction upon susceptible and oxidizable bonds.

We developed an efficient alternative method for the hydrofracturing process by using novel linear gels, prepared with biodiesel and water samples from different sources. Based on the improved results obtained in the pilot tests, several fluid systems were successfully implemented in various oil field operations. The study showed that using seawater with a gel made with biodiesel for hydrofracturing jobs is 80% more efficient and economical, as the gel needed no additional breakers. This method is novel, eco-friendly, sustainable, and advantageous as it saves freshwater usage. The gel viscosity was measured at various bottom hole-circulating temperatures in the range of100°F to 150°F. The breaker test was conducted at 113°F and 140°F in 120 min for the application of gel. Out of all the water samples, the sample collected from the produced water source was found to be more economical. It is feasible to reuse that water, which could help to minimize water usage. This approach is also environmentally friendly as it reduces produced water spills and contamination. The seawater-formulated fluids were found to be effective at different downhole temperatures and are compatible with a large variation of fracturing fluids and additives. Ultimately, the use of seawater lowered the operation cost by 30% to 40% of the overall operational cost.

This paper introduces to the industry a newly developed advanced Biodiesel gel that significantly enhances the oilwell recovery treatment efficiency, and it is a Novel, eco-friendly biodiesel linear gel. It also provides an explanation of the performed lab testing and the methodology of linear gel preparation.

Introduction

Biomass waste streams are a potential feedstock for a variety of eco-friendly products and serve as a crucial alternative energy source for achieving carbon neutrality. Consequently, the efficient management of biomass waste has gained increased importance in recent years (Basumatary S. et al., 2018). Due to its comparable physicochemical properties to fossil diesel, biodiesel is considered a promising substitute for conventional fossil fuels (Nath, B. et al., 2019).

This study aimed to synthesize biodiesel from the widely available, non-edible seed oil of sisymbrium irio. (a member of the Brassicaceae family) via a transesterification process using a homemade TiO_2 catalyst. A 93% biodiesel yield was achieved using a 1:16 oil-to-methanol ratio, 20 mg of catalyst, at 60 °C for 60 minutes. Fuel properties were analyzed according to ASTM methods (Naylor, R. L. et al., 2018).

Hydraulic fracturing involves pumping large volumes of fluid into oil or gas wells at high pressure to create fractures in rock formations, thereby facilitating the extraction of hydrocarbons (Hyne N. J., 2012). Traditionally, drinking water has been used as the primary fluid for hydraulic fracturing. However, this practice poses challenges such as water scarcity and contamination (Michael B. S. et al., 2015).

It is therefore important to develop hydraulic fracturing techniques that not only recover and reduce water contamination but also prevent microbial contamination (Chen W. et al., 2012). Recently, polymer gel-based fracturing fluids have been developed to address these issues. However, these gels are susceptible to enzymatic degradation by aerobic bacteria in the base water (Corrie E. C. et al., 2013). If uncontrolled, microbial growth can render the polymer non-functional. To mitigate this, biocides or bactericides are added to limit bacterial contamination (Bao Y. et al., 2016).

Currently, fossil diesel is commonly used in the preparation of both linear and cross-linked gels, which is environmentally harmful and unsustainable (James G. S., 2016). Using fossil diesel in fracturing fluids contributes to harmful carbon dioxide emissions, thereby exacerbating global warming (Vengosh A. et al., 2014). Studies suggest that vegetable-based biodiesel can serve as a viable alternative. Biodiesel synthesized from Jatropha seeds has been highlighted as a promising eco-friendly substitute (Wang T. et al., 2015).

Unconventional reservoirs, which typically exhibit low porosity and permeability, are difficult to exploit. Our previous work reported the use of basic hydrocarbon-derived linear gels for hydraulic fracturing in such formations (Wang Q. et al., 2006). Hydraulic fracturing is carried out by injecting these gels at high pressure and under bottom-hole static temperatures (Yew C. H. et al., 2015). A common base material in fracturing fluids is guar gum (Zhixi C. et al., 2006) (Fig. 1), which has a strong affinity for water and forms long-chain, high molecular weight polymer gels (Wang C. et al., 2017) (Fig. 2).

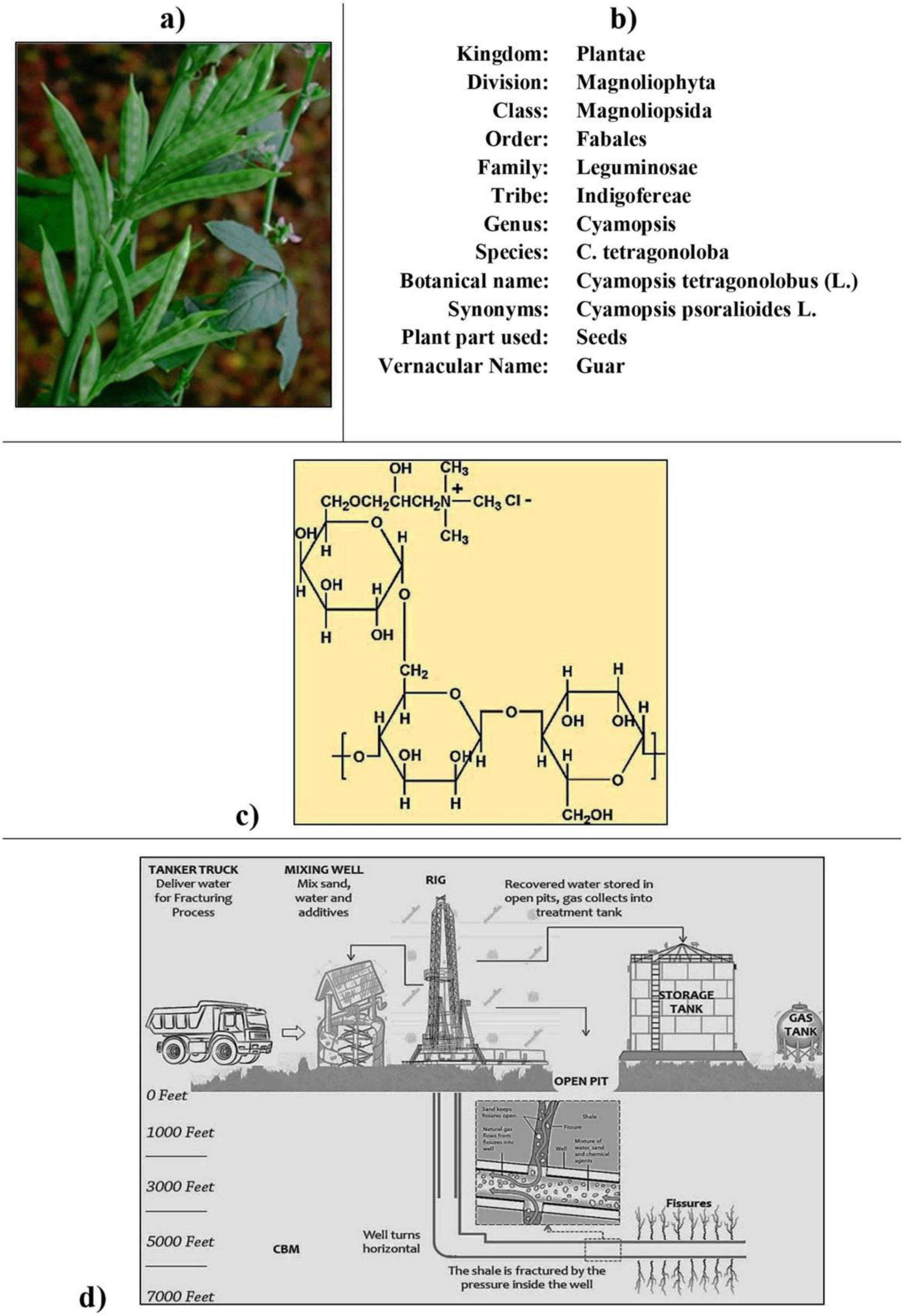

Figure 1—(a) Guar gum plant. (b) Guar scientific classification. (c) Chemical structure of guar gum. (d) Hydraulic fracturing process

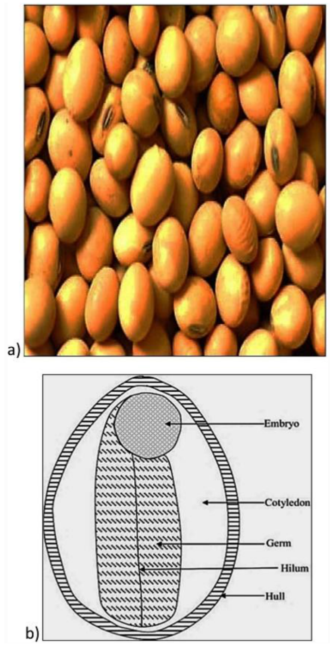

Figure 2—(a) Guar seed (b) Guar Seed anatomy

Guar gum, also known as guaran, is a galactomannan polysaccharide extracted from guar beans. It possesses thickening and stabilizing properties useful across food, cattle feed, and industrial applications (Xu L. et al., 2016; Huang Q. et al., 2019). Its derivatives, such as those composed of mannose and galactose, offer enhanced viscosity properties (Woodworth, T. R. et al., 2007; Siddharth J. et al., 2010). When hydrated, guar gum molecules unfold, significantly increasing the viscosity of the solution (Waters G. et al., 2009; Abdulmumin O. A. et al., 2016).

Hydraulic fracturing operations use diesel-powered pumps and engines to inject fluids into wells (Ramana Murthy R. V. et al., 2016; Yu W. et al., 2013). Biodiesel offers a cleaner burning alternative fuel that can reduce emissions from these engines (Stringfellow W. T. et al., 2017). Power at remote well sites is often generated using diesel generators, which can also run on biodiesel. Fleet vehicles involved in fracking operations, including trucks and transport vehicles, can utilize biodiesel blends (Xu B. et al., 2011). Additionally, biodiesel derivatives may serve as biodegradable lubricants or hydraulic fluids, though this application is less common than fuel usage (Zhang Q. B. et al., 2014).

Materials and methods

- Biodiesel, as shown in Figure 3, is a renewable, biodegradable fuel made from vegetable oils, animal fats, or recycled cooking oils. It is produced through a chemical process called transesterification, where triglycerides (fats/oils) react with an alcohol (usually methanol or ethanol) in the presence of a catalyst.
- Organophilic clay which is in solid state i.e. this clay gallant having enhanced dispersibility in organic systems, is prepared from the reaction product of a smectite type clay having a cation exchange capacity of at least 0.75 mill equivalents per gram and from 1.00 to less than 1.20 mill equivalents per gram of clay of a methyl benzyl dialkyl ammonium compound, wherein the compound contains 20 to 35% alkyl groups having 16 carbon atoms and 60 to 75% alkyl groups having 18 carbon atoms.
- Isotridecanol ethoxylated (anti-settling additive) is in liquid form.
- Sodium bicarbonate is a buffering agent for maintaining PH.

- Guar gum is a polymer.
- Two breakers, Ammonium per Sulphate, which is in a solid state, and another one is a liquid Hemicellulase enzyme.
- Gas flow which is in liquid state (composition of Methanol 2-Butoxyethanol Ethylene oxide-nonylphenolpolymer Alcohols, C12-16, and ethoxylated Tridecyl alcohol Nonylphenol Ethoxylated nonionic surfactant). It will flow back after frac job.
- Biocide is a composition formed from ingredients comprising peroxide and a hypochlorite.

Figure 3—Bio-diesel prepared by Transesterification

Experimental Studies and Results

Preparation of Biodiesel

Transesterification Method. A catalyst plays a major role for the preparation of Biodiesel. On the laboratory scale, dissolve NaOH in methanol. Mix about 3.5 g of NaOH with 200 mL of methanol. Stir well until fully dissolved. This forms sodium methoxide, a highly reactive solution. Warm the vegetable oil to about 55–60°C and slowly add the methoxide solution to the warmed oil while stirring continuously, then continue stirring the mixture for 1 to 2 hours. Let the mixture sit in a separatory funnel or container for several hours (typically overnight). I observed that two distinct layers will form. The Upper layer is Biodiesel (methyl ester), and the lower layer is glycerol (by-product). Carefully drain the glycerol from the bottom and collect the biodiesel layer. Then, wash the biodiesel with warm water to remove any residual methanol, catalyst, and soap. Gently mix with water and let it settle and repeat 2–3 times until the water becomes clear. Heat the biodiesel gently or allow it to sit in a warm, dry area to evaporate any remaining water. This glycerol can be developed as pharma grade glycerol, industrial grade glycerol, and cosmetic grade.

Chemical Reaction:

$$\text{Triglyceride}\left(\text{oil}/\text{fat}\right) + 3\text{Methanol} \xrightarrow{NaOH/KOH} 3\text{Methyl esters}\left(\text{Biodiesel}\right) + \text{Glycerol}$$

Synthesis of Bio-Diesel Hydro frac concentration for 500 ml XLFC-1B. Take 309 ml of Bio-Diesel into the Blender, should maintain the velocity at (1900-2000 rpm). Take 4.5 g of Organophilic mud (anti-settling agent), which contains minerals whose surfaces have been covered with a concoction to make them oil-dispersible powder, and add it to the Blender, mix for up to 10 minutes. This anti-settling agent must be allowed to disperse and yield. Take 2 ml of isotridecanol ethoxylated, which is high-performance, low-foaming, and highly biodegradable. It is a very good emulsifier and has good detergency, biodegradable, and therefore eco-friendly, which added to the Blender, agitate for 10 minutes in a liquid state. While taking this additive, it should be measured in a 5ml syringe. Add 60 g of Buffering agent like sodium Bicarbonate, which should be in powder state, mix until the material is dispersed and lump-free for 15 minutes. After mixing the buffering agent, the slurry should be maintained in a basic state. Need to check the basic nature by placing a pH strip into the slurry. Now add 240 g of guar gum and continue the mixing for 20 minutes (or) until the concentration of the slurry is smooth and lump-free. Here, the slurry obtained is cross-linked frac concentration as shown in Figure 4. Measure the density of the slurry. If the density of the slurry varies more than 0.1 ppg, refer to "cut back (or) weight-up" charts to correct the slurry density. Measure the viscosity of the slurry; it should be maintained at 250 cp in order to pump the slurry.

Figure 4—Bio-Diesel frac concentration XLFC-1B

Table 1—Bio-Diesel Hydro frac concentration slurry physical properties:

Specific Gravity	1.160
Slurry Density	9.750 ppg
Free Diesel	<2% (in 24 hrs)
Polymer Concentration	4.0 ppg
Flash Point	400°F (205°C)
Viscosity @ 300 RPM	280 cp

Procedure for preparation of 1000 ml gel Hydration. Consider 7.5 gpt Bio-Diesel frac concentrations, which means take 7.5 ml Frac concentration should mix with 1000 ml water sample, using a Chandler engineering high speed mixer at 2000 rpm, the gel should agitate for up to 3 min and stop the blender, and check the viscosity. First time we observed viscosity reading as 27 centipoises (cp). Again, agitate the gel up to 10 minutes, finally, we got the gel viscosity is 32 cp, the 7.5 gpt linear gel shown in Figure 5.

Figure 5—7.5 gpt Linear gel

Breaker Test. 1 ppt=1 pound/1000x1 gallon=1x453.2/1000x1x3.782=453.2/3782=0.12gms/lit.

Where 453.2 grams is a factor, 1 gallon = 3.782 liters.

1gpt =1-gallon per1000 gallons=1ltr/1000 ltr=1 ml/1000ml.

Then add 1ppt of ammonium persulfate and 1gpt of Enzyme-G into 32 cp Linear Gel glass bottle and mix it well. Put it in water baths that are maintained at 113°F and 140 °F temperatures, then check the viscosity every 10 minutes.

Test-1 Linear Gel prepared with Drinking water.

Table 2—Breaker Test (7.5 gpt gel under breaker concentration at 1 ppt oxidizer and 1 gpt Enzyme-G).

S. No	Gel Breaking Time (minutes)	Linear gel viscosity (cp) 1 ppt & 1 gpt breakers	
		113°F	140°F
1	0	32 cp	32 cp
2	10	31	32
3	20	30	31
4	30	27	30
5	40	24	28
6	50	22	25
7	60	21	23
8	90	19	21
9	120	15	19
10	150	13	17
11	180	10	16

Test-2 Linear Gel prepared with sea water.

Table 3—Breaker Test (7.5 gpt gel) without Breaker

S. No	Gel Breaking Time (minutes)	Linear gel viscosity (cp)	
		113°F	140°F
1	0	32 cp	32 cp
2	10	28	29
3	20	26	27
4	30	23	24
5	40	20	21
6	50	18	19
7	60	15	16
8	90	12	13
9	120	10	11
10	150	8	9
11	180	5	6

Conclusion

Biodiesel is a clean, renewable alternative to fossil diesel, offering environmental and economic benefits. The most common production method is base-catalyzed transesterification. It is a renewable, biodegradable fuel made from vegetable oils, animal fats, or recycled cooking grease. It offers several advantages over conventional diesel. It is environmentally friendly and reduces greenhouse gases (CO_2, SO_2), particulate matter, and hydrocarbon emissions compared to petroleum diesel. Biodegradable Breaks down faster in the environment, reducing pollution risks, non-toxic, less harmful to humans and wildlife. It is a renewable & sustainable composition made from renewable sources like soybean oil, canola oil, palm oil, or waste cooking oil, and reduces dependence on fossil fuels. This research explored the development of alternative methods for the hydro fracturing process, which are feasible and eco-friendly. Pilot tests were also conducted for the first time by using biodiesel as the alternate fuel and were found to be more economical. The successful execution of pilot tests of the frac operations and the fruitful results are detailed. When compared with fossil diesel XLFC, the biodiesel XLFC shows better and improved results for on-shore and off-shore fracturing operations. When compared with fossil Diesel, XLFC Biodiesel gives good results for frac operations and is economically less expensive. The study found that seawater-formulated fluids are effective across a range of downhole temperatures and are compatible with various fracturing fluids and additives. A key benefit highlighted is that the use of seawater led to a 30% to 40% reduction in overall operational costs, making it a cost-effective alternative in hydraulic fracturing operations.

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